Experiment 9: Rule-Based Part-of-Speech Tagging in Python

1. Aim

To write a Python program to implement a rule-based part-of-speech tagging system using regular expressions.

2. Procedure

1. Import `RegexpTagger` from the `nltk.tag` module.
2. Define a list of regular expression rules mapping suffix and number patterns to POS tags.
3. Initialize the `RegexpTagger` with the defined rule patterns.
4. Apply `tagger.tag()` on a tokenized test sentence.
5. Display the rule-based POS tagged output.

3. Program

```
from nltk.tag import RegexpTagger

patterns = [(r'.*ing$', 'VBG'), (r'.*ed$', 'VBD'), (r'.*ly$', 'RB'), (r'.*s$', 'NNS'), (r'^[0-9]+$', 'CD'), (r'.*', 'NN')]

tagger = RegexpTagger(patterns)

words = ["dogs", "barked", "loudly", "running", "3", "cat"]
print("Rule-Based Tags:", tagger.tag(words))
```

4. Output

```
Rule-Based Tags: [('dogs', 'NNS'), ('barked', 'VBD'), ('loudly', 'RB'), ('running', 'VBG'), ('3', 'CD'), ('cat', 'NN')]
```

5. Result

Thus, the Python program was successfully executed to implement a rule-based part-of-speech tagging system using `RegexpTagger`.